# Lecture 27 – Generators, Motors and Associated Gizmos

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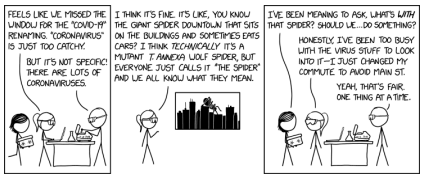

# 1 Eddy Currents

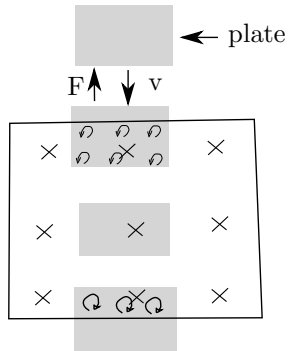


Figure 1: Plate and field

Force will oppose velocity, so this results in a concept known as 'magnetic braking', in which metallic objects are slowed while passing thhrough a magnetic field.

# 2 Generators & Motors

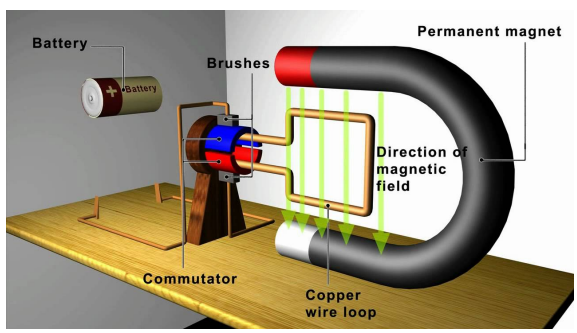


Figure 2: Electromagnetic Generator

- Produces an alternating current (AC) when turned.
- With a battery, can be made to work in reverse – by giving a battery with electric potential V, can result in kinetic energy.

# 3 AC vs DC

- Main electricity supply of the US is 120V, and oscillates at 50-60Hz.
- Historic debate between these two, with a rivalry between Nikola Tesla and Thomas Edison, representing AC and DC respectively.
- AC:
  - Lightbulbs produc more present light (?).
  - Transporting AC current is more efficient than DC.
- DC:
  - All microchips require DC.
  - Nicer equation!

## 3.1 Transformer

Method to convert between different AC voltages.

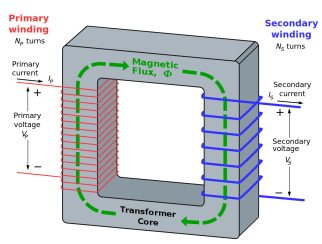


Figure 3: Transformer diagram

$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

Electrical pylons carry around 15,000 Volts, but household supplies require only around 120V.
Converted through these transformers – same power, but different end voltage.
Can convert between AC and DC.

### 3.2 Diode bridge

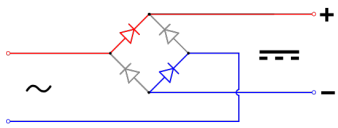


Figure 4: Diode bridge